

# SAGE-RAI: Design Patterns for Transparent RAG Systems

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## Abstract

Retrieval-Augmented Generation (RAG) systems are increasingly deployed in web-based educational environments, yet transparency can be seen as a primarily ethical, and, too often, optional, concern, rather than foundational. This paper presents design patterns for building transparent RAG systems, derived from developing and deploying SAGE-RAI, an advanced multi-purpose RAG system, in an educational context. Through systematic evaluation combining quantitative rating data ( $n=26$ , mean rating=4.62/5) and qualitative interviews ( $n=4$ ), we demonstrate that transparency serves dual pedagogical and ethical functions. Our empirical findings reveal high user satisfaction (92.3% rating 4-5 stars) while identifying critical tensions between AI assistance and learning independence. Our findings suggest that as RAG systems increasingly mediate access to web-based knowledge, transparency must evolve from an optional feature to an architectural requirement.

## CCS Concepts

• Information systems → Question answering; • Computing methodologies → Natural language generation.

## Keywords

Retrieval-Augmented Generation (RAG), Transparency, Responsible Web, Educational AI, Design Patterns

### ACM Reference Format:

Joseph Kwarteng, Aisling Third, Alexander Mikroyannidis, David Tarrant, and John Domingue. 2026. SAGE-RAI: Design Patterns for Transparent RAG Systems. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3774904.3792929>

## 1 Introduction

The web is undergoing a fundamental transformation in how users access information. Large Language Models (LLMs) are rapidly becoming primary interfaces for web-based knowledge, with educational institutions increasingly embedding AI assistants to support learning tasks [23]. Yet, this shift introduces critical challenges:

when AI mediates access to web information, how do we ensure learners develop necessary critical evaluation skills for web literacy? How do we prevent algorithmic authority from replacing critical thinking? How do we handle sensitive topics when content safety filters may inappropriately restrict legitimate educational materials [3, 19]? Not forgetting the risks of plausible but incorrect responses; “hallucination” [15, 30]. Retrieval-Augmented Generation (RAG) mitigates some risks by grounding generation in retrieved web sources [20], but often, systems treat transparency (e.g., revealing sources, reasoning processes, and system mechanics) as optional rather than foundational. This opacity (“black box”) risks creating a culture where learners accept AI outputs uncritically, undermining the web’s potential as an educational resource.

We argue that transparency in educational RAG is not just an ethical requirement but a pedagogical necessity. When learners can inspect sources, verify claims, and understand how knowledge is assembled from web sources, transparency can transform a trust mechanism into a learning scaffold. However, technical complexity and lack of established best practice can prevent educators and domain experts from using them most effectively to support learning.

This paper presents design patterns derived from developing SAGE-RAI (Smart Assessment and Guided Education with Responsible AI), a multi-purpose RAG system evaluated through educational deployment. We focus not on RAG technicalities or general explainable AI, but specifically on transparency in online educational interactions. These patterns represent our empirically-grounded proposals for further validation in diverse contexts. Our findings have implications beyond education: as RAG systems increasingly mediate web access across domains, understanding effective transparency is crucial to maintain the web’s epistemic integrity.

**Contributions:** i) Design patterns for transparency in educational RAG, addressing when, where, and how to expose components without cognitive overload, ii) Empirical evidence of dual pedagogical and ethical functions of transparency in web-mediated learning, iii) An open-source reference implementation for replication and extension, available at (<https://github.com/SAGE-RAI>).

## 2 Related Work

**RAG Systems and Transparency:** Lewis et al. [20] showed that RAG improves factuality by grounding outputs in retrieved documents, sparking frameworks like LangChain and LlamaIndex. While advances in self reflective [2] and active retrieval [16] improve RAG performance, Gao et al. [11] reveal that these assume technical



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ACM ISBN 979-8-4007-2307-0/2026/04  
<https://doi.org/10.1145/3774904.3792929>

audiences, with citation as optional. Bohnet et al. [5] show fine-grained citations improve trust and verifiability, while Liu et al. [22] demonstrate that trust is eroded more by inaccurate citations than by none. We position source transparency as a core architectural requirement, examining the pedagogy of giving citations.

**Explainability and Trust:** XAI literature distinguishes *interpretable* and *actionable* [8, 21]. From HCI, transparency must provide usable information [17], echoing human-centred design, where system behaviours must be comprehensible to users [9]. However, this is complex: explanations can be ignored [24] or paradoxically increase reliance on incorrect outputs [29]. We instead examine transparency as pedagogical scaffolding to support critical thinking.

**Educational AI and Learning Independence:** Bloom's two-sigma finding [4] motivated extensive intelligent tutoring research [7, 28]. Recent work examines generative AI in education; Holstein et al. [13] identified tensions between AI assistance and learning independence, and Kasneci et al. [18] surveyed opportunities and risks. Transparency and its effect on learning behaviours are understudied, with most research focusing on learning outcomes instead. We contribute empirical evidence on transparency supporting learning.

**Responsible AI (RAI) in Education:** Gebru et al.'s datasheets [12] and Mitchell et al.'s model cards [25] provide AI documentation developer frameworks. Content filtering poses educational challenges, as overly restrictive filters may block legitimate content [3, 19]. We operationalize RAI principles for educational RAG, bridging the gap between technical and educational AI research through design patterns empirically grounded from learning perspectives.

### 3 Towards Transparent AI: Design Patterns

We present design patterns addressing **what** to make transparent and **when, how** to present it, and **why**, framed as problem–context–solution relationships generalisable beyond SAGE-RAI [1, 10].

#### Architectural Foundations

**P1: Modular Transparency Layers.** *Problem:* Users' (e.g., learners, educators, administrators) diverse transparency needs. *Solution:* Transparency layers: Interactional (citations, responses), Operational (prompts, models, knowledge base), Governance (privacy, data retention). *Rationale:* Overwhelming users can impair comprehension [27]; layered disclosure enables progressive access. *Implementation:* Learners see citations, prompts, knowledge base; educators, configuration dashboards; admins, governance dashboards.

**P2: Provider-Agnostic Architecture.** *Problem:* Institutions have diverse trust preferences and resource constraints. *Solution:* Seamlessly support hosted (OpenAI, Anthropic, Azure) and local deployment (Ollama, HuggingFace). *Rationale:* Institutional model choice without system lock-in. *Implementation:* Abstraction layers enable model switching and offer institutional data control.

**P3: No-Code Configuration.** *Problem:* Educators have pedagogical, not technical skills. *Solution:* Graphical interfaces for all configurations (knowledge base, prompts, models, privacy). *Rationale:* Democratizing AI requires removing technical barriers. *Implementation:* Web interface with drag-and-drop and bulk uploads (data ingestion), text-area prompt editing, and checkbox privacy controls.

#### Retrieval and Citation Patterns

**P4: Multi-Level Source Attribution.** *Problem:* Document-level citations prevent efficient claim verification. *Solution:* Maintain provenance through chunking, embedding, retrieval, generation; cite at chunk-level with verification context. *Rationale:* Fine-grained attribution enables efficient verification [5]. *Implementation:* Identify highest query-chunk relevance. Metadata includes source, chunk, page content, URL, timestamp; citations link directly to documents.

**P5: Dynamic Retrieval with Transparency.** *Problem:* Fixed retrieval strategies waste resources or risk hallucination. *Solution:* LLM assesses whether existing context suffices and logs decision. *Rationale:* Adaptive retrieval improves efficiency [16]; transparency gives explanations. *Implementation:* System determines when knowledge base retrieval needed; users see decisions & new sources.

**P6: Citation Verification Affordances.** *Problem:* Citations must be verifiable. *Solution:* Hyperlinked citations, showing source content in-context. *Rationale:* Reducing verification friction increases engagement. *Implementation:* Links open full sources in new tabs.

#### Knowledge Base Transparency

**P7: Inspectable Knowledge Base.** *Problem:* Users do not know AI response provenance. *Solution:* Visible sources and metadata. *Rationale:* Understanding scope enables appropriate queries and critical trust. *Implementation:* Searchable up-to-date sources/metadata.

**P8: Query Resolution Transparency.** *Problem:* Ambiguous queries (pronouns, references) are not understood by users. *Solution:* Expand queries for users from conversation history. *Rationale:* Explicit interpretation teaches metacognitive skills. *Implementation:* System expands follow-up references, e.g., "the event" to "the ODI Summit".

#### Interaction and Feedback Patterns

**P9: Uncertainty Communication.** *Problem:* False confidence encourages overreliance. *Solution:* Explicitly communicate uncertainty and knowledge gaps. *Rationale:* Calibrated confidence builds appropriate trust; admitting limitations is pedagogically superior to false certainty. *Implementation:* Default prompt enforces transparency on uncertainty, e.g., "Sources do not directly address this...".

**P10: Human-in-the-Loop.** *Problem:* Cannot always adequately answer queries. *Solution:* Explicit escalation to human instructors. *Rationale:* AI augments rather than replaces human judgment. *Implementation:* "Email tutor" when the system determines inadequacy or learners are unsatisfied; sends full conversation to educators.

**P11: Bidirectional Feedback.** *Problem:* Feedback loop is not closed. *Solution:* Enable ratings & feedback to educators. *Rationale:* Participatory design teaches agency in shaping technology. *Implementation:* Configurable ratings (1-5 stars) with optional (open) feedback & educator dashboard; students receive usage token compensation.

#### Privacy and Governance Patterns

**P12: Transparent Data Governance.** *Problem:* Users do not understand data collection, use, or access. *Solution:* Clear, accessible governance documentation. *Rationale:* Informed consent requires understandable information. *Implementation:* Plain-language privacy tools; links to relevant terms of service and user data policies.

## 4 Evaluation & Findings

### 4.1 Methodology

We evaluated SAGE-RAI via mixed-methods combining quantitative ratings (n=26) and qualitative interviews (n=4) 10-weeks in a Data Ethics Professional course at the Open Data Institute.

**Deployment Context** SAGE-RAI was configured as a course-specific assistant with a knowledge base of all course materials (lectures, readings, resources), available 24/7. Students received guidance on appropriate use (complementing rather than replacing engagement with materials). Our design patterns were implemented using Meta's Llama 3 for generation, nomic-embed [26] for semantic retrieval, and hosted in the UK and EU for GDPR compliance.

**Data Collection:** Students could optionally rate responses (1-5 stars) with text feedback through the integrated system (P11). Four volunteers (3 female, 1 male) from diverse professional (government, non-profit, water management, technology) and educational backgrounds (law, business, geography, data science) participated in 40-60 minute semi-structured interviews, covering: (1) usage patterns, (2) perceived benefits and limitations, (3) transparency, (4) ethical concerns, and (5) suggestions for improvement. Transcripts were analyzed thematically, following Braun and Clarke [6].

**Ethical Considerations:** Institutional ethical approval was received. Participation did not affect grades, and volunteers gave informed consent. Data was stored securely and anonymized.

### 4.2 Quantitative Findings:

We collected 26 ratings over the 10-week study, from students voluntarily rating their interactions with SAGE-RAI. While this is a subset of usage (not all interactions were rated), the ratings provide valuable insights into user satisfaction and system performance.

**Overall Satisfaction:** The ratings demonstrated strong overall satisfaction. The mean rating of 4.62 (SD=0.64) and median of 5.0 indicate consistently positive user experiences. Notably, 92.3% of ratings were high (4-5 stars), with 69.2% awarding the maximum 5 stars. Only 7.7% of ratings were moderate (3 stars), and no ratings were below 3. This distribution suggests the transparent design successfully supported learner needs while maintaining trust.

**Feedback Message Analysis:** Of the 26 ratings, 11 (42.3%) included optional text feedback, providing qualitative context for the scores. The majority of feedback messages (9 of 11) were explicitly positive, using terms like "Excellent response" (5 instances), "Completely correct" (2 instances), and 1 instance each of "Mostly correct" and "Very helpful". These aligned with 4-5 star ratings, suggesting that the transparent design, particularly source attribution and uncertainty communication, helped users verify response quality. One 3-star rating included valuable constructive feedback: *"Misinterpreted what I was asking. It thought I was asking about data ethics professionals in general, not the course entitled Data Ethics Professionals. I still got the info I needed from the sources though."* This comment is particularly instructive. While it identifies a query resolution failure (P8), the student's ability to recognize the misinterpretation and still extract useful information from the provided sources shows the value of transparency. The source citations (P6) enabled the student to work around the misunderstanding, which validates our argument that transparency serves as a safety mechanism against AI errors.

### 4.3 Qualitative Findings

Thematic analysis revealed three interconnected themes that triangulate with quantitative findings, explaining how transparency features enabled high satisfaction while surfacing critical tensions.

**Transparency-Enabled Opportunities:** Participants valued SAGE-RAI as a *"critical friend"*, enhancing learning through quick content access, assignment validation, and criteria checking. Transparency features were central: *"...I still got the info I needed from the sources,"* illustrates how they enabled productive engagement even when responses missed the mark. Participants emphasized the complementary role of the assistant, allowing tutors to focus on deeper, more personalized guidance, validating P10 (human-in-the-loop). Source verification empowered learners to test their understanding without fear while maintaining critical evaluation capacity.

**Limitations as Boundary Awareness:** Participants identified over-generality ("too broad/high-level; needs follow-up prompts") and scope constraints ("limited to course material; lacks external/current context"), but framed these as boundary awareness, not failures; transparency of knowledge sources (P7) tempered expectations. Concerns of over-reliance "reducing deeper research" show metacognitive awareness, suggesting transparency prompted reflection on appropriate use rather than uncritical acceptance.

**Ethical Sophistication:** Participants distinguished appropriate use (validating understanding) from misuse (plagiarism), enabled by transparency, when sources are visible. They emphasized "AI should assist rather than replace educators" and called for "strong guardrails," with institutional support and pedagogical framing.

## 5 Discussion

Our evaluation shows that systematic transparency serves dual pedagogical and ethical functions, with implications extending beyond education to broader web-mediated knowledge access.

**Transparency as Pedagogical Infrastructure:** Our findings validate Kaminski's [17] argument that transparency must be meaningful and actionable. Transparency enabled learners to: (1) verify claims against sources, maintain critical thinking, and productively work around errors, supporting findings on fine-grained attribution [5]; (2) calibrate trust, addressing overreliance concerns [29]; (3) escalate to human expertise, operationalizing teacher-AI complementarity [13]; and (4) reflect metacognitively, evidenced by participants' ethical sophistication.

High satisfaction despite acknowledged limitations suggests transparency enables productive engagement within system boundaries. Could this go contrary to XAI research [24] suggesting users ignore explanations in educational contexts? With pedagogical framing, learners actively engage with transparency as learning scaffold rather than a mere trust signal.

**Resolving Critical Tensions:** Three tensions extend theoretical debates in educational AI. The *accessibility-detail tension* reflects cognitive load theory [27]: learners value both concise summaries and comprehensive detail. Our layered transparency (P1) addresses this through progressive disclosure, aligning with Doshi-Velez and Kim's [8] context-appropriate interpretability. The *scope-breadth tension* validates Liu et al.'s [22] finding that citation quality affects trust: constraining knowledge bases improves attribution but reduces breadth. This suggests hybrid architectures with

transparent mode, switching between “course-specific” and “web-augmented” retrieval, extending Gao et al.’s [11] RAG frameworks. The *assistance-independence tension* engages Holstein et al.’s [13] identification of tensions between AI support and learning autonomy. Transparency makes assistance visible rather than “magical”, but consistent with Kasneci et al.’s [18] analysis, resolution requires pedagogical framing, assessment design, and institutional policies beyond technical design.

**Implications for Responsible Web:** Emerging technologies such as Model Context Protocol (MCP) [14], and RAG systems increasingly mediate web knowledge, creating new provenance and traceability challenges. Architectural transparency must preserve the web’s critical evaluation structures [12, 25]. Our patterns demonstrate systematic implementation: layered disclosure (P1) prevents cognitive overload, provider-agnostic architecture (P2) supports institutional autonomy, and multi-level attribution (P4) preserves citation networks—applicable to journalism, *chunk-level citations* (P4, P7) *facilitate fact-checking*; healthcare, *uncertainty communication* (P9) *supports clinical decision-making*; and legal domains, (P2) *ensures confidentiality*. However, transparency alone is insufficient [17]. Responsible Web requires ecosystems supporting critical engagement: digital literacy, institutional support, and accountability mechanisms addressing algorithmic bias [3, 19].

**Limitations and Future Directions:** Our modest scale (26 ratings, 4 interviews) from one professional course limits generalizability. These patterns require validation across diverse contexts. Voluntary ratings may introduce selection bias, and without learning outcomes or engagement logs, we cannot determine which patterns are functionally used. Future work should: (1) experimentally compare outcomes with/without transparency, (2) investigate cross-disciplinary and cultural variations, (3) examine high-stakes domains, and (4) validate patterns across multiple implementations.

## 6 Conclusion

This paper demonstrates that transparency in educational RAG systems is an architectural requirement, not optional. We contribute twelve empirically grounded design patterns that implement transparency without cognitive overload (layered disclosure), enable source verification (multi-level attribution), and maintain human-AI complementarity (escalation mechanisms). However, our findings also revealed that transparency alone is insufficient; it requires pedagogical framing and institutional support to resolve tensions between AI assistance and learning independence. As RAG systems increasingly mediate web-based knowledge access, systematic transparency can preserve the web’s epistemic integrity, enabling critical engagement rather than passive consumption. SAGE-RAI’s open-source implementation invites the community to extend and validate these patterns across diverse contexts, building AI-mediated knowledge access transparently, accountably, and in service of human learning.

## Acknowledgments

SAGE-RAI funded by RAi UK (UKRI ref. EP/Y009800/1), grant IA069.

## References

- [1] Christopher Alexander, Sara Ishikawa, and Murray Silverstein. 1977. *A Pattern Language: Towns, Buildings, Construction*. Oxford University Press, New York.

- [2] Akari Asai, Zequi Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection.
- [3] Ryan S Baker and Aaron Hawn. 2022. Algorithmic bias in education. *Int. journal of artificial intelligence in education* 32, 4 (2022), 1052–1092.
- [4] Benjamin Bloom. 1984. The 2-sigma Problem. *Educ. Researcher* 13, 6 (1984), 4–16.
- [5] Bernd Bohnet, Vinh Q Tran, Pat Verga, Roei Aharoni, Daniel Andor, Livio Baldini Soares, Massimiliano Ciaramita, Jacob Eisenstein, Kuzman Ganchev, Jonathan Herzig, et al. 2022. Attributed question answering: Evaluation and modeling for attributed large language models. *arXiv:2212.08037* (2022).
- [6] Victoria Clarke, Virginia Braun, and Nikki Hayfield. 2015. Thematic analysis. *Qualitative psychology: A practical guide to research methods* 3 (2015), 222–248.
- [7] Cristina Conati, Kaska Porayska-Pomsta, and Manolis Mavrikis. 2018. AI in Education needs interpretable machine learning. *arXiv:1807.00154* (2018).
- [8] Finale Doshi-Velez and Been Kim. 2017. Towards a rigorous science of interpretable machine learning. *arXiv:1702.08608* (2017).
- [9] Upol Ehsan and Mark O Riedl. 2020. Human-centered explainable ai: Towards a reflective sociotechnical approach. In *International Conference on Human-Computer Interaction*. Springer, 449–466.
- [10] Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides. 1994. *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley, Boston.
- [11] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2024. Retrieval-Augmented Generation for Large Language Models. *arXiv:2312.10997*
- [12] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé Iii, and Kate Crawford. 2021. Datasheets for datasets. *Commun. ACM* 64, 12 (2021), 86–92.
- [13] Kenneth Holstein, Bruce M McLaren, and Vincent Aleven. 2019. Co-designing a real-time classroom orchestration tool to support teacher-AI complementarity. *Grantee Submission* (2019).
- [14] Xinyi Hou, Yanjie Zhao, Shengao Wang, and Haoyu Wang. [n. d.]. Model Context Protocol (MCP). *arXiv:2503.23278* ([n. d.]).
- [15] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of Hallucination in Natural Language Generation. *Comput. Surveys* 55, 12 (March 2023), 1–38.
- [16] Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. Active retrieval augmented generation. In *Proc. of the 2023 Conf. on Empirical Methods in NLP*. 7969–7992.
- [17] Margot E Kaminski. 2021. The right to explanation, explained. In *Research handbook on information law and governance*. Edward Elgar Publishing, 278–299.
- [18] Enkeleida Kasneci, Kathrin Seßler, Stefan Küchemann, Maria Bannert, Daryna Dementieva, Frank Fischer, Urs Gasser, Georg Groh, Stephan Günnemann, Eyke Hüllermeier, et al. 2023. ChatGPT for good? On opportunities and challenges of LLMs for education. *Learning and individual differences* 103 (2023), 102274.
- [19] Joseph Kwarteng, Serena Coppolino Perfumi, Tracie Farrell, Aisling Third, and Miriam Fernandez. 2022. Misogynoir: challenges in detecting intersectional hate. *Social Network Analysis and Mining* 12, 1 (2022), 166.
- [20] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2021. RAG for Knowledge-Intensive NLP Tasks. *arXiv:2005.11401*
- [21] Pantelis Linardatos, Vasilis Papastefanopoulos, and Sotiris Kotsiantis. 2020. Explainable AI: A review of machine learning interpretability methods. *Entropy* 23, 1 (2020), 18.
- [22] Nelson F Liu, Tianyi Zhang, and Percy Liang. 2023. Evaluating verifiability in generative search engines. *arXiv:2304.09848* (2023).
- [23] Alexander Mikroyannidis, Audrey Ekuban, Joseph Kwarteng, and John Domingue. 2025. Best Practices for the Responsible Adoption of Generative AI in Higher Education. In *Proceedings*, Vol. 114. MDPI, 6.
- [24] Martijn Millecamp, Nyi Nyi Htun, Cristina Conati, and Katrien Verbert. 2019. To explain or not to explain: In *Proc. of the 24th Int. Conf. on Intelligent User Interfaces*. 397–407.
- [25] Margaret Mitchell, Simone Wu, Andrew Zaldivar, Parker Barnes, Lucy Vasserman, Ben Hutchinson, Elena Spitzer, Inioluwa Deborah Raji, and Timnit Gebru. 2019. Model cards for model reporting. In *Proc. of the conf. on fairness, accountability, and transparency*. 220–229.
- [26] Zach Nussbaum, John X. Morris, Brandon Duderstadt, and Andriy Mulyar. 2025. Nomic Embed: Training a Reproducible Long Context Text Embedder.
- [27] John Sweller. 2011. Cognitive load theory. In *Psychology of learning and motivation*. Vol. 55. Elsevier, 37–76.
- [28] Kurt VanLehn. 2011. The relative effectiveness of human tutoring, intelligent tutoring systems & other tutoring systems. *Educ. Psychol.* 46, 4 (2011), 197–221.
- [29] John Zerilli, Alistair Knott, James Maclaurin, and Colin Gavaghan. 2019. Transparency in algorithmic and human decision-making: is there a double standard? *Philosophy & Technology* 32, 4 (2019), 661–683.
- [30] Yue Zhang, Yafu Li, Leyang Cui, Deng Cai, Lemao Liu, Tingchen Fu, Xinting Huang, Enbo Zhao, Yu Zhang, Chen Xu, Yulong Chen, Longyue Wang, Anh Tuan Luu, Wei Bi, Freda Shi, and Shuming Shi. 2025. Siren’s Song in the AI Ocean: A Survey on Hallucination in Large Language Models. *arXiv:2309.01219*